

WEEK - 3 PRACTICAL

The screenshot shows the MongoDB Atlas Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar lists the database structure: Cluster0, admin, bookflow, book_metadata, local, and studentDB. The main panel displays the 'book_metadata' collection with a single document. The document details are as follows:

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e80'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

At the bottom, the 'Alert System Status' is 'All Good'.

This screenshot shows the same MongoDB Atlas Data Explorer interface, but with a list of documents displayed. The 'Documents' tab shows 5 documents. The first document is expanded, showing the following details:

```
{
  "_id": ObjectId('6a9143ae6d442b6cd5cf3e86'),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 105,
  "reviews": Array (2)
}
```

The interface also shows a search bar and navigation controls. The 'Alert System Status' at the bottom remains 'All Good'.

cloud.mongodb.com/v2/6a8280b1669cc9eb12eca55d#/explorer/6a8281ded5972af80226fe6c/bookflow/book_metadata/find

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - local
 - studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

25 1 - 5 of 5

```
member_id: 101
rating: 5
comments: "Excellent science fiction story"
1: Object (3)
  member_id: 102
  rating: 4
  comments: "Very interesting and informative"
2: Object (3)
  member_id: 103
  rating: 5
  comments: "Excellent space adventure"
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a8280b1669cc9eb12eca55d#/explorer/6a8281ded5972af80226fe6c/bookflow/book_metadata/find

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Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

25 1 - 5 of 5

```
1: Object (3)
  member_id: 101
  rating: 5
  comments: "Excellent science fiction story"
2: Object (3)
  member_id: 102
  rating: 4
  comments: "Very interesting and informative"
3: Object (3)
  member_id: 103
  rating: 5
  comments: "Excellent space adventure"
```

Alert System Status: All Good

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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

25 1 - 4 of 4

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e80')
  book_id: 101
  title: "The Martian"
  author: "Andy Weir"
  published_year: 2021
  category: "Science Fiction"
  format: "Digital PDF"
  file_size: "2.5 MB"
  member_id: 101
}
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a8280b1669cc9eb12eca55d#/explorer/6a8281ded5972af80226fe6c/bookflow/book_metadata/aggregation

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Cluster0 > bookflow > book_metadata

Documents 5 **Aggregations** Schema Indexes 1 Validation

View monitoring Visualize Your Data

\$match

Untitled - modified + CREATE NEW

PREVIEW

Stage 1 | \$match

```
1 {
2   "published_year": {
3     "$gt": 2020
4   }
5 }
```

Output preview after \$match stage (Sample of 4 documents)

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e80')
  book_id: 101
  title: "The Martian"
  author: "Andy Weir"
  published_year: 2021
  category: "Science Fiction"
  format: "Digital PDF"
  file_size: "2.5 MB"
  member_id: 101
}
```

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Data Explorer

My Queries Data Modeling

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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match \$unwind

Run Options

Stage 2 \$unwind

```
1 /**
2  * path: Path to the array field.
3  * includeArrayIndex: Optional name for
4  * preserveNullAndEmptyArrays: Optional
5  * toggle to unwind null and empty val
6  */
7 {
8   path: "$reviews"
9 }
10
```

Output preview after \$unwind stage (Sample of 10 documents)

```
{
  "_id": ObjectId("6a9143ae6d442b6cd5cf3e86"),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 185
}
```

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cloud.mongodb.com/v2/6a8280b1669cc9eb12eca55d#/explorer/6a8281ded5972af80226fe6c/bookflow/book_metadata/aggregation

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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match \$unwind \$group

Run Options

Stage 3 \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
{
  "_id": "Data Structures",
  "average_rating": 4,
  "total_reviews": 2
}
```

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Documents 5 Aggregations Schema Indexes 1 Validation

\$match \$unwind \$group \$sort

Untitled - modified + CREATE NEW

PREVIEW

Stage 4 \$sort

```
1 |
2 {
3   "average_rating": -1
4 }
```

Output preview after \$sort stage (Sample of 4 documents)

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```

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cloud.mongodb.com/v2/6a8280b1669cc9eb12eca55d#/explorer/6a8281ded5972af80226fe6c/bookflow/book_metadata/indexes

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Documents 5 Aggregations Schema Indexes 2 Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	4 (since Fri Aug 28 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Fri Aug 28 2026)		READY

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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

View monitoring Visualize Your Data

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

25 1 - 5 of 5

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e80'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

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ORGANIZATION PROJECT

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree Raw Output

```
{
  "parsedTextQuery": {
    "terms": ["excel"],
    "negatedTerms": [],
    "phrases": [],
    "negatedPhrases": []
  },
  "textIndexVersion": 3,
  "inputStage": {
    "stage": "FETCH",
    "inputStage": {
      "stage": "IXSCAN",
      "keyPattern": {
        "_fts": "text",
        "_ftsx": 1
      },
      "indexName": "reviews.comments_text",
      "isMultiKey": true,
      "isUnique": false,
      "isSparse": false,
      "isPartial": false,
      "indexVersion": 2,
      "key": {
        "_fts": "text",
        "_ftsx": 1
      }
    }
  }
}
```

Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:

- _fts (text)
- _ftsx ↑

Close